

Stanford University
Political Science 350D
Problem Set 1
Due April 6, 2015

1. In a study of teen recidivism, there were $n = 43$ persons released from incarceration, of whom $y = 15$ were re-offenders within 36 months. We wish to estimate the probability θ of recidivism.
 - (a) Using a Uniform(0, 1) prior for θ , plot the prior, likelihood, and posterior distributions of θ . Find the posterior mean, median, mode and standard deviation of θ and a 95% credible interval for θ .
 - (b) Repeat (a), but use a Beta(2, 8) prior for θ .
 - (c) Repeat (a), but use a Beta(8, 2) prior for θ .

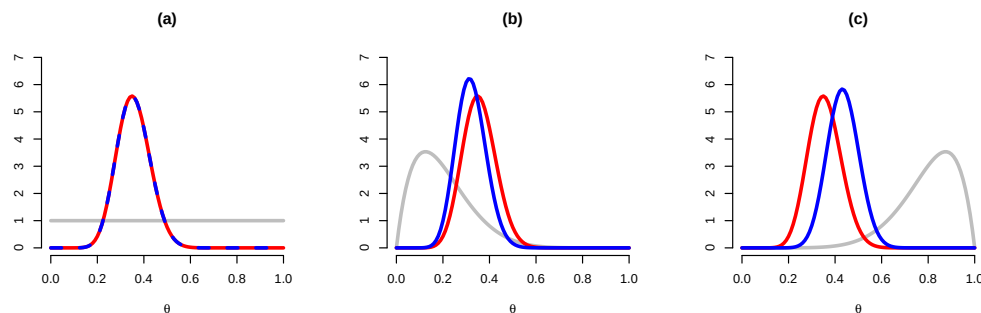
Solution: This problem uses some properties of the Beta distribution. If $\theta \sim \text{Beta}(a, b)$, then

- Mean: $\mathbb{E}(\theta) = a/(a + b)$
- Variance: $\mathbb{V}(\theta) = ab/(a + b)^2(a + b + 1)$
- Mode: $(a - 1)/(a + b - 2)$ (Differentiate $\theta^{a-1}(1 - \theta)^{b-1}$ with respect to θ .)

For $\theta \sim \text{Beta}(a, b)$, the median and 95% credible intervals can be obtained using R:

```
qbeta(1/2, shape1=a, shape2=b)
qbeta(c(.025, .975), shape1=a, shape2=b)
```

In parts (a)—(c), the prior distributions are all Beta(a, b) distributions with parameters (a) $a = b = 1$, (b) $a = 2, b = 8$, and (c) $a = 8, b = 2$. The posterior distribution has parameters $a + y$ and $b + n - y$. With these priors, the posteriors are Beta($a + y, b + n - y$) for the different choices of a and b in the prior. Note that the likelihood function is always the same, regardless of the prior.



In the graph, the prior is shown in gray, the (normalized) likelihood in red, and the posterior in blue. The likelihood $p(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$ is “normalized” by dividing by $\int_0^1 p(y|\theta) d\theta = \binom{n}{y} B(y + 1, n - y + 1)$.

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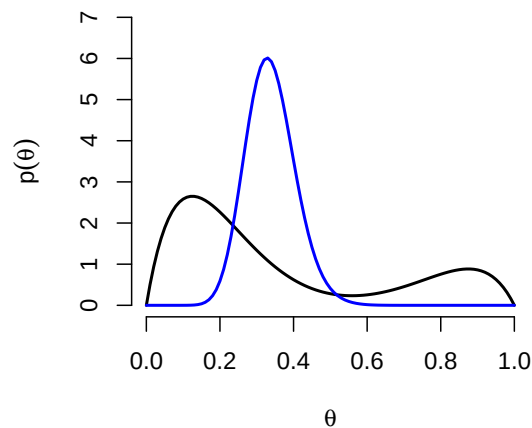
	Prior	Mean	Median	Mode	SD	95% C.I.
(a)	Uniform(0, 1)	0.356	0.353	0.349	0.071	(0.224, 0.499)
(b)	Beta(2, 8)	0.321	0.318	0.314	0.064	(0.203, 0.451)
(c)	Beta(8, 2)	0.434	0.433	0.431	0.067	(0.305, 0.568)

(d) Consider the following prior distribution for θ :

$$p(\theta) = \frac{1}{4} \frac{\Gamma(10)}{\Gamma(2)\Gamma(8)} (3\theta(1-\theta)^7 + \theta^7(1-\theta)),$$

which is a 75-25 percent mixture of a Beta(2, 8) and a Beta(8, 2) prior distribution. Plot this prior distribution and compare it to the priors in (b) and (c). Describe what sort of prior opinion this may represent.

Solution: The prior is a mixture of a Beta(2, 8) prior (with $\mathbb{E}(\theta) = 0.2$) and a Beta(8, 2) prior (with $\mathbb{E}(\theta) = 0.8$), with probability 3/4 placed on the former and 1/4 on the latter. This prior is appropriate when there is a strong division of opinion or two conceptually distinct models with diametrically opposed predictions, *e.g.*, most juvenile offenders can be rehabilitated by incarceration vs. most juvenile offenders are hardened criminals and incarcerating them only makes it worse. The 75-25 weighting means that we believe the former model is three times as plausible as the latter model.



(e) For the prior in part (d):

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- i. Write an expression for $p(\theta)p(y|\theta)$ and simplify as much as possible.

Solution:

$$\begin{aligned} p(\theta) p(y|\theta) &= \frac{1}{4B(2, 8)} (3\theta(1 - \theta)^7 + \theta^7(1 - \theta)) \binom{43}{15} \theta^{15}(1 - \theta)^{43-15} \\ &= \frac{1}{B(2, 8)} \binom{43}{15} \left(\frac{3}{4} \theta^{16}(1 - \theta)^{35} + \frac{1}{4} \theta^{22}(1 - \theta)^{29} \right) \\ &\propto \frac{3}{4} \theta^{17}(1 - \theta)^{35} + \frac{1}{4} \theta^{22}(1 - \theta)^{29} \end{aligned}$$

- ii. The posterior distribution is a mixture of two distributions you should know. Identify these distributions from their kernels.

Solution: The posterior is a 75-25 mixture of a $\text{Beta}(23, 30)$ distribution (with kernel $\theta^{18-1}(1 - \theta)^{36-1}$) and a $\text{Beta}(x, y)$ distribution (with kernel $\theta^{23-1}(1 - \theta)^{30-1}$).

- iii. Use R to calculate and plot the unnormalized posterior $p(\theta)p(y|\theta)$ by evaluating it on a grid of points. Identify the (approximate) posterior mode and compare to your answers in parts (a–c).

Solution: Approximately 0.33, which is close to the posterior obtained by using a $\text{Beta}(2, 8)$ prior.

```
> theta <- seq(0, 1, by=.01)
> posterior <- (3/4) * theta^17 * (1-theta)^35 +
+   (1/4) * theta^22 * (1-theta)^29
> theta[which.max(posterior)]      # posterior mode
[1] 0.33
```

- (f) Find a general formula for the weights on the mixture distribution in (d)(ii) and provide an interpretation for these values.

2. Install and use JAGS to answer Question 1(b).

Solution:

```
> library(rjags)
> model <- "model {
```

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```
+ y ~ dbin(theta, n)
+ theta ~ dbeta(2,8)
+ }"
> data <- list(y=15, n=43)
> sampler <- jags.model(textConnection(model), data=data)
Compiling model graph
  Resolving undeclared variables
  Allocating nodes
  Graph Size: 5

Initializing model

> samples <- jags.samples(sampler, variable.names=c("theta"),
+   n.iter=5000)
  |*****| 100%
> mean(samples$theta); median(samples$theta); sd(samples$theta)
[1] 0.3214275
[1] 0.319086
[1] 0.06382877
> quantile(samples$theta, probs=c(.025,.975))
      2.5%      97.5%
0.2026275 0.4523197
```

